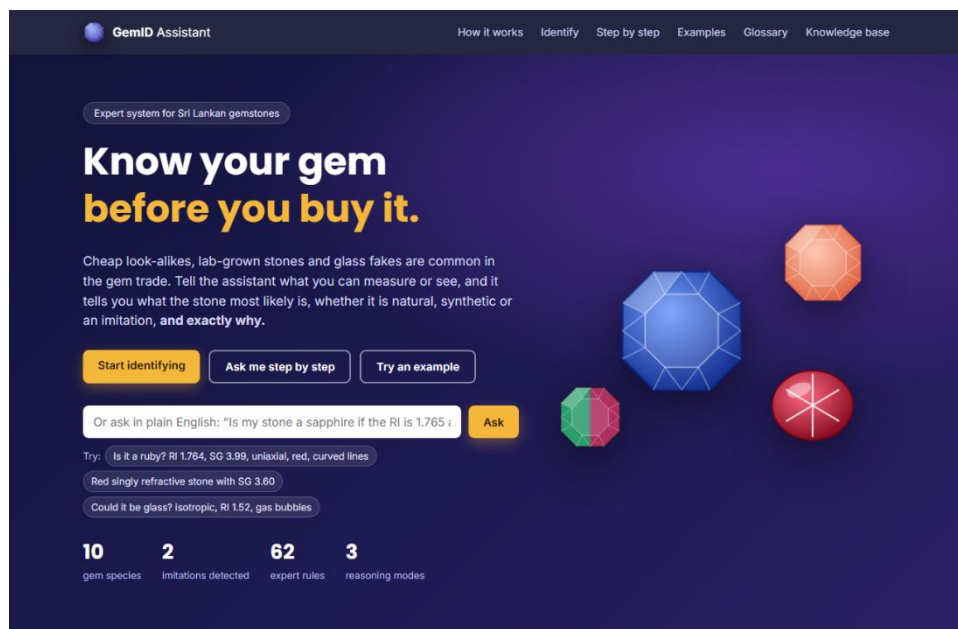


Gemstone Identification Assistant

User Manual

Installation, operation and worked examples



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Version 1.0 - 29 September 2026

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1. About the system

The Gemstone Identification Assistant is an expert system that helps buyers, students and small traders identify common Sri Lankan gemstones. You describe a stone (its measurements and what you can see) and the system tells you:

- what the stone most likely is (species and variety, for example *corundum* - *blue sapphire*);
- whether it is likely natural, synthetic (laboratory-grown) or an imitation such as glass;
- how certain it is, as a percentage;
- what you should do next, for example obtain a laboratory certificate before buying;
- exactly how it reached its answer.

It knows ten gem species (corundum, spinel, garnet, chrysoberyl, zircon, topaz, tourmaline, quartz, feldspar/moonstone and beryl), their main varieties, and two imitation materials (glass and cubic zirconia).

Important: the system gives a first opinion only. It does not replace a certificate from the National Gem and Jewellery Authority (NGJA) or another accredited gem laboratory. Always obtain a certificate before a significant purchase.

1.1 Three ways to use it

Mode	Use it when...	What it does
Identify	"I don't know what this stone is."	Takes everything you enter and works forwards to the most likely answer (forward chaining).
Verify	"The seller says it's a ruby. Is it?"	Starts from the claim and checks every condition it needs; if the claim fails it tells you exactly why (backward chaining).
Step by step	"I want to be guided."	Asks you one question at a time, only when needed, and explains why it is asking.

You can also type a question in plain English, for example *"Is my stone a sapphire if the RI is 1.765 and it is blue?"*

2. What you need

Table 1. Requirements

Item	Requirement
Computer	Windows 10 or 11, macOS or Linux, with about 100 MB of free disk space.
SWI-Prolog	Version 9 or later (free). The system was developed and tested with SWI-Prolog 10.1.16 (64-bit).
Web browser	Any modern browser: Microsoft Edge, Google Chrome, Firefox or Safari.
Project files	The <code>gem_expert</code> folder.

Internet	Only for downloading SWI-Prolog. The system itself runs offline; without internet the page simply uses standard fonts.
Gem instruments (optional)	Refractometer, hydrostatic balance, polariscope and a 10x loupe give the best results, but the system also works with only what you can see.

3. Installation

3.1 Install SWI-Prolog

1. Open <https://www.swi-prolog.org/download/stable> in your browser.
2. Download the installer for your computer. On Windows choose the **64-bit** installer (`swipl-...-1.x64.exe`).
3. Run the installer and accept the default settings.
4. **Windows only:** on the "Install Options" page select **"Add swipl to the system PATH for all users"** (or "for the current user"). This lets you start SWI-Prolog from a terminal.
5. Finish the installation. On macOS you can alternatively install with Homebrew (`brew install swi-prolog`); on Ubuntu Linux use `sudo apt install swi-prolog`.

[Insert a screenshot of the SWI-Prolog installer's "Add swipl to the system PATH" option here.]

3.2 Copy the project folder

1. Copy or unzip the `gem_expert` folder to a convenient place, for example `Documents\gem_expert`.
2. Check that the folder contains these files: `main.pl`, `knowledge_base.pl`, `engine.pl`, `ui.pl`, `console.pl`, `diagrams.pl`, `examples.pl`, `query_parser.pl`, `tests.pl` and an `images` folder.

3.3 Check the installation

1. Open a terminal: on Windows, open the `gem_expert` folder in File Explorer, click the address bar, type `cmd` and press Enter. On macOS or Linux, open Terminal and use `cd` to go to the folder.
2. Type `swipl --version` and press Enter. You should see a line such as `SWI-Prolog version 10.1.16 for x64-win64`.

If you see "'swipl' is not recognized", SWI-Prolog is not on the PATH. See Troubleshooting (Section 12).

4. Starting and stopping the system

4.1 Start the web interface (recommended)

1. In the terminal, inside the `gem_expert` folder, type:

```
swipl main.pl
```

You will see:

```
=== Gemstone Identification Assistant loaded ===  
server(8080).    start web interface, then open http://localhost:8080  
start.          console interface  
demo.           worked example  
run_tests.      run automated tests  
  
?-
```

2. At the `?-` prompt type the following, **including the full stop**, and press Enter:

```
server(8080).
```

The system replies `Web interface running: open http://localhost:8080 in your browser.`

3. Open your web browser and go to **`http://localhost:8080`**. The home page appears (Figure 1).

On Windows the first time you start the server, Windows Defender Firewall may ask whether to allow SWI-Prolog to communicate. Choose **Allow** for private networks. The system only listens on your own computer.

Every Prolog command ends with a full stop. If nothing happens after pressing Enter, type `.` and press Enter again.

4.2 Stop the system

1. In the terminal type `stop_server(8080).` to stop the web server.
2. Type `halt.` to leave SWI-Prolog. (Closing the terminal window also stops everything.)

5. A tour of the home page

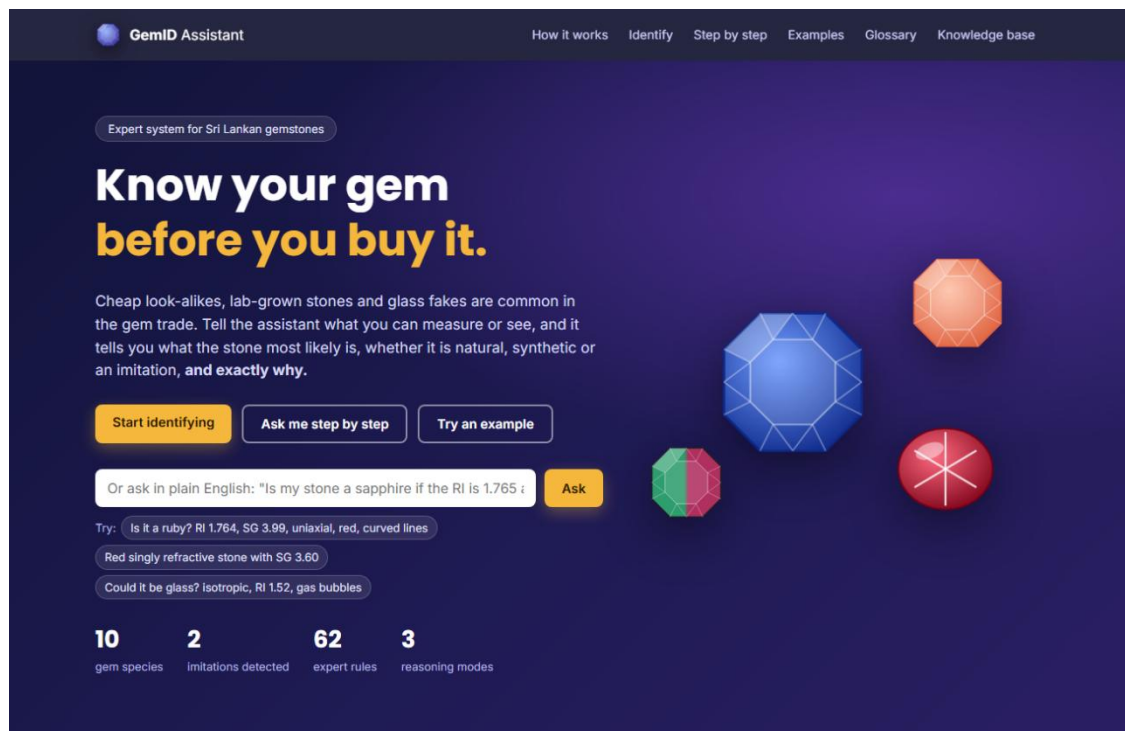


Figure 1. Home page

The top bar links to every part of the system: **How it works**, **Identify** (the input form), **Step by step**, **Examples**, **Glossary** and **Knowledge base**. The introduction offers three buttons: **Start identifying**, **Ask me step by step** and **Try an example**, plus a box for plain-English questions. Scrolling down you will find a summary of what the system tells you, an explanation of how it works and of the two reasoning methods (Figure 2), the input form, example stones and a glossary of gem terms.

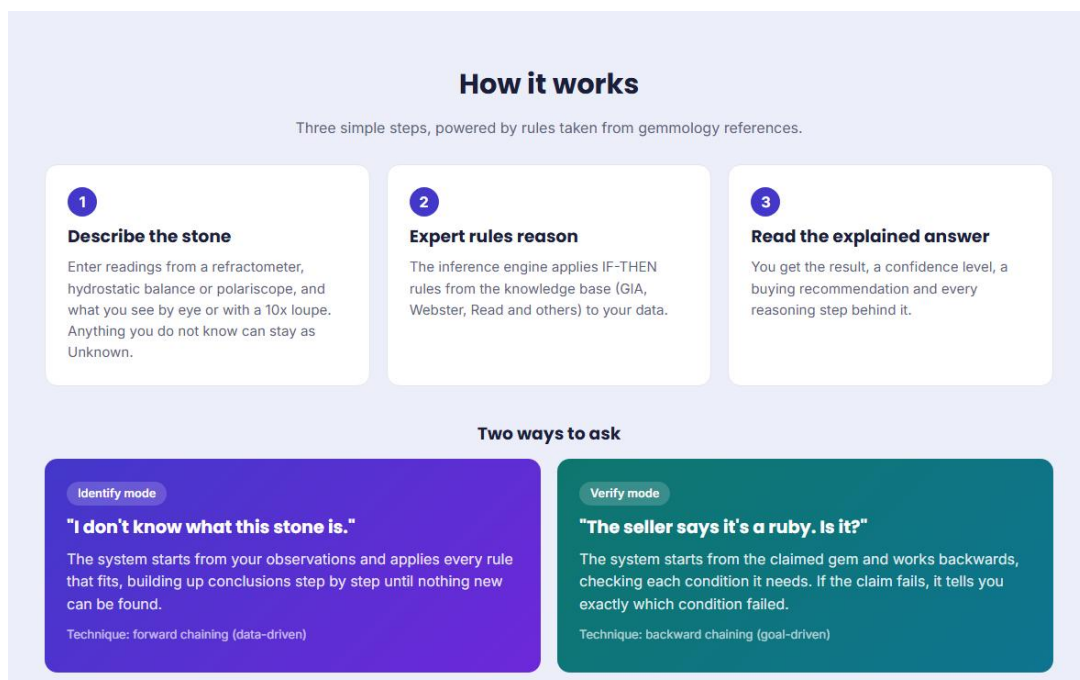


Figure 2. "How it works" section

6. Describing your stone

The form (Figure 3) has two groups of fields. **Every field is optional.** Fill in what you know and leave the rest as *Unknown / not checked*. The more you fill in, the more certain the answer. Each field shows what the property means (grey text) and how to measure it (**HOW**).

Describe your stone

Every field is optional. The more you fill in, the more confident the answer.

1 Instrument readings

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)
How heavy the stone is compared with the same volume of water.

HOW Weigh the stone in air and in water (hydrostatic balance).

Optic character
Whether light passes through as one ray or is split in two.

HOW Turn the stone between crossed filters in a polariscope.

2 What you can see

Colour
The main body colour of the stone.

HOW Look at it in daylight against a white background.

Optical effect
Special light effects such as a star or a cat's-eye.

HOW Look under a single light source and rotate the stone.

Inclusions
Tiny features inside the stone that reveal how it formed.

HOW Examine it with a 10x jeweller's loupe.

Identify my stone
Find the best match using every rule in the knowledge base.

Check a claim
Test whether the stone could be one particular gem.

Ask me step by step
The system asks only the questions it needs, and you can ask it why.

[Clear the form](#)

Example stones to try

Figure 3. The input form

Table 2. Input fields

Field	What it is	How to obtain it	Tips
Refractive index (RI)	How strongly the stone bends light. Each gem has its own range.	Gem refractometer with contact liquid; use the highest reading.	Enter three decimals, e.g. 1.765 . Allowed range 1.3-3.0. Leave blank if the scale shows no edge (over the limit) or you have no refractometer.
Specific gravity (SG)	How heavy the stone is compared with the same volume of water.	Hydrostatic balance: SG = weight in air / (weight in air - weight in water).	Enter two decimals, e.g. 4.00 . Allowed range 1.0-8.0.
Optic character	Whether light passes as one ray (singly refractive) or is split in two (doubly refractive).	Polariscope: rotate the stone between crossed filters. A conoscope shows uniaxial or biaxial.	If you only know it is doubly refractive, choose <i>Doubly refractive (axis not known)</i> .

Colour	The main body colour.	Look in daylight against a white background.	Choose the closest colour.
Optical effect	Star, cat's-eye, colour change or moonstone glow.	Look under a single light source and rotate the stone; compare daylight and lamp light for colour change.	Choose <i>No special effect</i> if you looked and saw none.
Inclusions	Tiny features inside the stone that reveal how it formed.	Examine with a 10x jeweller's loupe.	Choose the most obvious type. <i>None visible</i> means you looked and the stone is clean.

Below the fields are three action panels: **Identify my stone**, **Check a claim** (choose a gem in the list, then **Verify**) and **Ask me step by step**. **Clear the form** empties all fields.

7. Identifying a stone

7.1 Steps

1. Fill in the fields you know (Section 6).
2. Press **Identify stone**.
3. Read the result page (Figure 4).

7.2 Reading the result

The screenshot shows a web interface for gemstone identification. At the top, there's a navigation bar with links: Assistant, How it works, Identify, Step by step, Examples, Glossary, and Know. Below this, a 'Back to home' link is visible. The main heading is 'Your result' in a large, bold font. Underneath, it says 'Here is what the expert system concluded from your data.' The central part of the page features a large white box with a dark blue background on the left containing a 3D model of a blue sapphire. To the right of the model, the text reads 'MOST LIKELY Blue sapphire' in bold. Below this, it says 'Gem species: Corundum'. There are two green status bars: 'Strong match - 90% certain' and 'Likely natural (76%)'. A 'Certainty' slider is shown at 90%. Below this box, there's a section titled 'What you should do' with a green arrow icon and the text: 'Features are consistent with a natural stone. Obtain an NGJA (National Gem and Jewellery Authority) or other accredited laboratory certificate before purchase.' At the bottom, a section titled 'What you told us' displays a series of buttons representing the user's input: 'RI 1.765', 'SG 4.0', 'Optic Uniaxial', 'Colour Blue', 'Effect not given', and 'Inclusions Silk'.

Figure 4. Result for a natural blue sapphire

Table 3. Parts of the result page

Part	Meaning
Title (e.g. <i>Blue sapphire</i>)	The most likely identification, with the species underneath.
Confidence badge	Strong match (green): optic character, RI and SG all agree. Possible match (amber): data is incomplete or fits more than one gem. No match (grey): nothing in the knowledge base fits.
Certainty bar	How strongly the expert rules believe the conclusion, from 0% to 100%.
Origin badge	Likely natural (green), Synthetic or laboratory test recommended (amber), Imitation or Glass or synthetic (red), Not determined (grey, usually because no inclusions were reported). A Probably heat-treated badge appears when heat indicators were seen.
What you should do	Practical recommendations.
What you told us	Your inputs; fields you left empty show "not given".
Reasoning diagram	A picture of how the rules fired (Section 7.3).
How the system reached this	Each reasoning step in plain English, with the rule number, its certainty, the expert knowledge behind it and its source.
About (gem)	Reference facts about the identified gem.
Technical trace	Click to open the full list of facts and fired rules (for gemmologists and examiners).
Buttons	Edit my inputs returns to the form with your values filled in; Start a new stone clears it.

7.3 Using the reasoning diagram

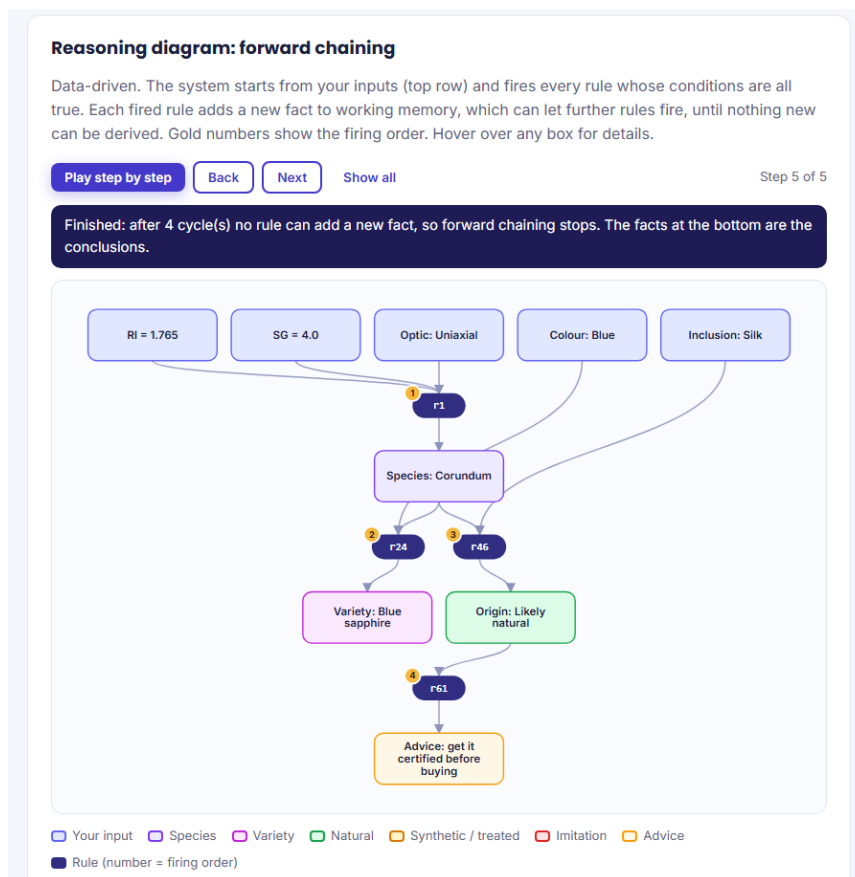


Figure 5. Forward-chaining diagram

Your inputs are the blue boxes at the top. Each dark oval is a rule; its gold number shows the order in which it fired. Arrows show which facts made the rule fire and which new fact it added. Use the buttons:

- **Play step by step:** replays the reasoning automatically, one rule at a time, with a caption explaining each step.
- **Back / Next:** move one step at a time yourself.
- **Show all:** shows the complete diagram again.

Hover the mouse over any box to see its full details.

7.4 When some data is missing

If you leave out the RI, SG or optic character, the system cannot confirm a species. Instead it lists **Ranked possibilities** (Figure 6): every gem consistent with what you entered, ordered by certainty, with the evidence for each. Measure the missing properties to narrow the list.

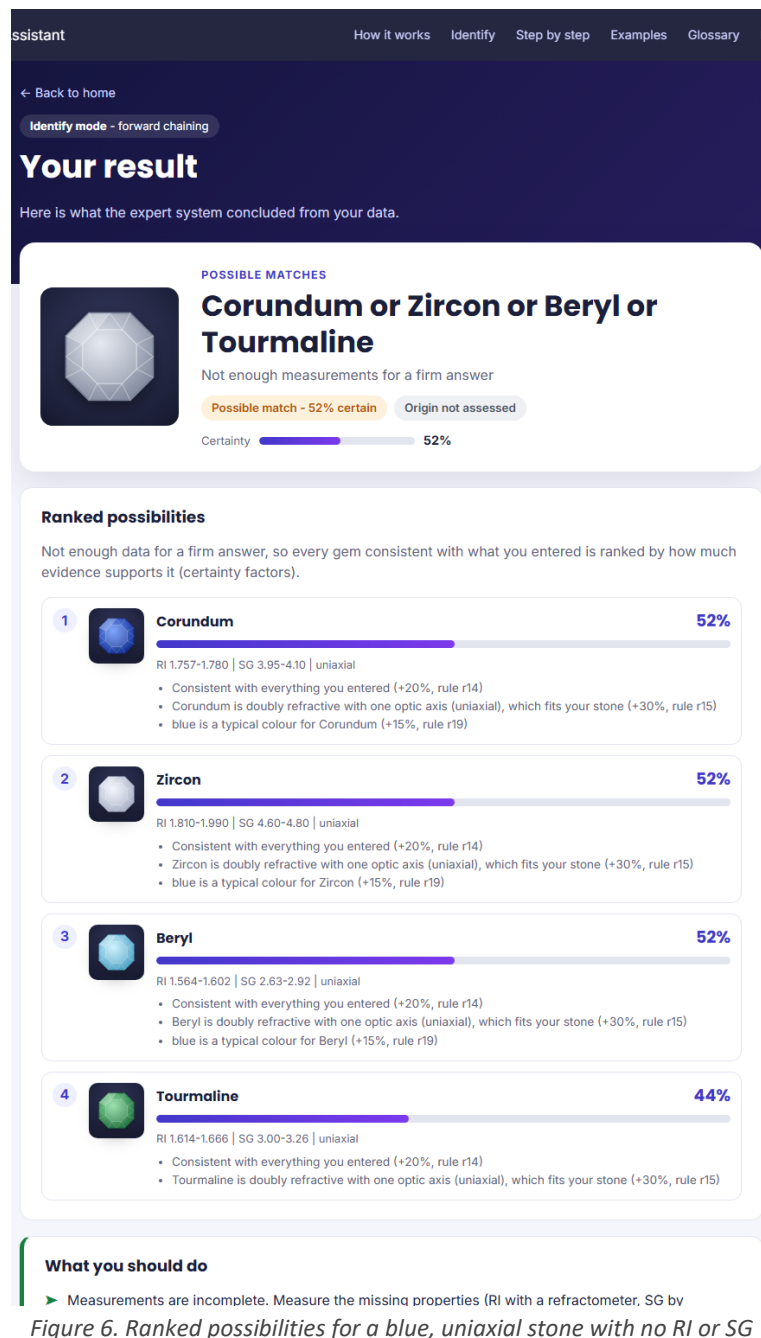


Figure 6. Ranked possibilities for a blue, uniaxial stone with no RI or SG

8. Checking a claim (Verify)

1. Fill in the fields you know.
2. In the **Check a claim** panel choose the gem the seller claims, e.g. *Ruby*.
3. Press **Verify**.

The answer is a clear **Yes** (green tick) or **No** (red cross). For a **No**, the page shows the **main reason**, what your data points to instead, and a **Why not?** list tracing the failure back through the rules (Figure 7). The backward-chaining diagram shows the question at the top, the rules tried below it, and each condition marked proved (green), failed (red) or never checked (dashed grey).

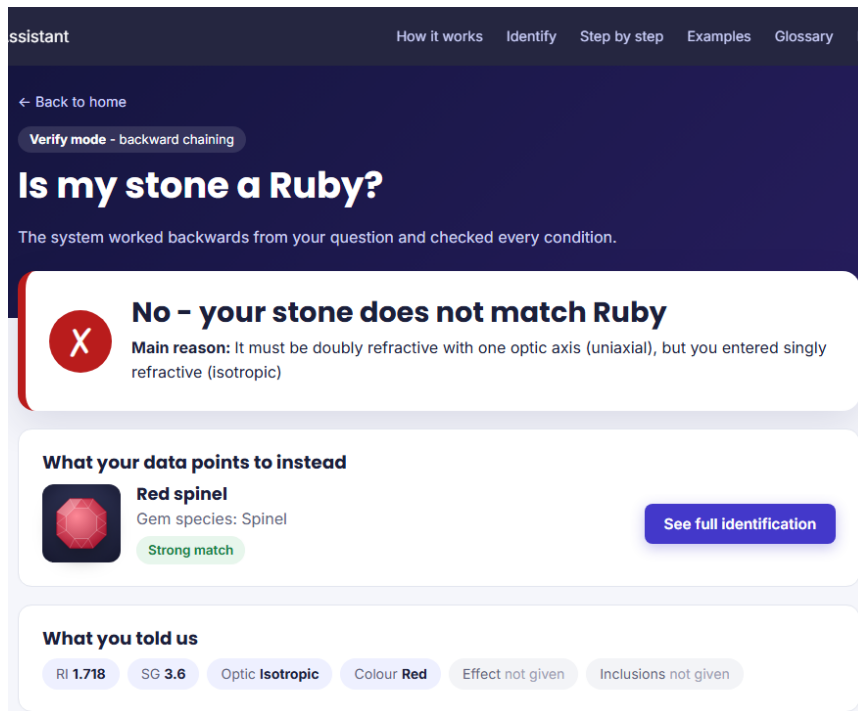


Figure 7. A failed verification: a red spinel offered as a ruby

9. Step-by-step consultation

1. Click **Step by step** in the top bar (or **Ask me step by step** on the home page).
2. Choose what you want to find out. *Find the species* lets the system test one gem hypothesis after another. You can also choose a particular gem.
3. Answer each question and press **Answer**, or press **I don't know**.
4. Open **Why are you asking this?** at any time to see which rule the system is testing and why it needs the answer (Figure 8).
5. When the system has enough information it shows the answer, how many questions it needed, and a diagram of its reasoning.

ssistant How it works Identify Step by step Examples Glossary K

← Back to home

Consult mode - Interactive backward chaining

Is my stone a Ruby?

The system tests one hypothesis at a time and asks only for the facts the current rule needs.

QUESTION 1 testing rule r1

What is the optic character of your stone?

Optic character
Whether light passes through as one ray or is split in two.

Unknown / not checked

HOW Turn the stone between crossed filters in a polariscope.

Answer I don't know

▼ Why are you asking this?

1. I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red AND NOT Effect: Star THEN Variety: Ruby.
2. To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
3. To test rule r1 I need to know: the optic character.

[Start the consultation again](#)

Figure 8. A consultation question with its explanation

The system asks only what it needs. For example, when checking for ruby, answering *Isotropic* to the first question ends the consultation immediately, because a ruby must be doubly refractive.

10. Asking in plain English

Type a question into the box on the home page and press **Ask**. The system shows what it understood (Figure 9) and then answers. Examples:

- *Is my stone a sapphire if the RI is 1.765 and it is blue?*
- *What is a singly refractive red stone with SG 3.60?*
- *Could it be glass? isotropic, RI 1.52, gas bubbles*
- *RI 1.52, SG 2.57, biaxial, colourless, with adularescence*

Understood words include RI / refractive index, SG / specific gravity, singly or doubly refractive, uniaxial, biaxial, colour names, star, cat's eye, colour change, adularescence, silk, crystals, fingerprints, gas bubbles, curved lines, swirl marks, discoid fractures, clean, and gem names. Words the system did not use are listed so you can rephrase.

Assistant

How it works Identify Step by step Examples Glossary Knowledge

← Back to home

Verify mode - backward chaining

Is my stone a Blue sapphire?

The system worked backwards from your question and checked every condition.

You asked

"Is my stone a sapphire if the RI is 1.765 and it is blue?"

The grammar (a Prolog DCG) understood your question as:

Question **Is it Blue sapphire?** RI **1.765** Colour **Blue**

No - your stone does not match Blue sapphire

Main reason: It must be doubly refractive with one optic axis (uniaxial), but the optic character was not given

What your data points to instead

Corundum

Not enough measurements for a firm answer

Possible match

See full identification

What you told us

Figure 9. A plain-English question and how it was understood

11. Worked examples

The **Example stones to try** section of the home page has 14 ready-made stones (Figure 10). Press **Try it** on any card to copy its readings into the form, then press **Identify stone** or **Verify**.

example stones to try

Press "Try it" to load a stone's readings into the form, then choose Identify or Verify.

Blue sapphire RI 1.765 SG 4.00 Optic Uniaxial Colour Blue Inclusion Silk EXPECTED Blue sapphire, likely natural Try it	Synthetic ruby RI 1.764 SG 3.99 Optic Uniaxial Colour Red Inclusion Curved growth lines EXPECTED Ruby, synthetic (flame fusion) Try it	Star sapphire RI 1.765 SG 4.00 Optic Uniaxial Colour Blue Effect Star (asterism) Inclusion Silk EXPECTED Star sapphire, likely natural Try it	Padparadscha RI 1.766 SG 4.00 Optic Uniaxial Colour Orange Inclusion Fingerprints / healed fractures EXPECTED Padparadscha, likely natural Try it
---	---	--	--

Figure 10. Example stones

11.1 Example 1: a natural blue sapphire

Input	RI 1.765, SG 4.00, Uniaxial, Blue, Silk
-------	---

Result	Blue sapphire (species corundum) - Strong match, 90% certain; origin Likely natural (76%) .
Recommendation	Obtain an NGJA or other accredited laboratory certificate before purchase.
Why	Rule r1 identified corundum from the optic character, RI and SG; rule r24 named the blue variety; rule r46 used the rutile silk as evidence of natural origin.

11.2 Example 2: a glass imitation

Input	RI 1.60, Isotropic, Round gas bubbles (SG not measured)
Result	Imitation: Glass - 85% certain; origin Imitation (not a natural gemstone) (Figure 11).
Recommendation	This is an imitation with little gem value; do not pay gemstone prices.

The screenshot shows a web application interface for gemstone identification. At the top, there's a navigation bar with links like 'How it works', 'Identify', 'Step by step', 'Examples', 'Glossary', and 'Knowledge base'. Below this, a 'Back to home' link is visible. The main heading is 'Your result' with a subtext 'Here is what the expert system concluded from your data.' The result section features a blue octagonal icon with a geometric pattern. To its right, it says 'IMITATION DETECTED' in blue, followed by 'Glass' in large bold text. Below this, it states 'This is not a natural gemstone'. There are two horizontal bars: a green one labeled 'Strong match - 85% certain' and a red one labeled 'Imitation (not a natural gemstone) (85%)'. A 'Certainty' slider is shown at 85%. Below the result, a section titled 'What you should do' contains a green arrow icon and the text 'This is an imitation, not a natural gemstone. It has little gem value - do not pay gemstone prices.' Another section titled 'What you told us' lists input data: 'RI 1.6', 'SG not given', 'Optic Isotropic', 'Colour not given', 'Effect not given', and 'Inclusions Round gas bubbles'. At the bottom, it says 'Reasoning diagram: forward chaining'.

Figure 11. A glass imitation is detected

11.3 Example 3: checking a claim that fails

Input	RI 1.718, SG 3.60, Isotropic, Red; claim: <i>Ruby</i>
Result	No - the stone does not match ruby. Main reason: a ruby must be doubly refractive (uniaxial) but the stone is singly refractive. The data points to Red spinel instead (Figure 7).

12. Troubleshooting

Table 4. Common problems and solutions

Problem	Solution
"swipl" is not recognized as an internal or external command"	SWI-Prolog is not on the PATH. Re-run the installer and select "Add swipl to the system PATH", then open a new terminal. Alternatively run it with its full path: "C:\Program Files\swipl\bin\swipl.exe" main.pl.
Error "port 8080 is already in use" (or similar) after <code>server(8080).</code>	Another program uses that port. Start on a different port, e.g. <code>server(8081).</code> , and open <code>http://localhost:8081</code> .
Browser shows "This site can't be reached"	The server is not running. Check that you typed <code>server(8080).</code> with the full stop, and that the terminal window is still open. Check the port number in the address.
Nothing happens after typing a command	Prolog is waiting for the full stop. Type <code>.</code> and press Enter.
Error messages about files not found when running <code>swipl main.pl</code>	The terminal is not in the <code>gem_expert</code> folder. Use <code>cd</code> to move into it, or open the terminal from that folder.
"Please check your input" page	A value is outside the allowed range (RI 1.3-3.0, SG 1.0-8.0) or is not a number. Press Go back and fix it .
Result says "We could not identify this stone"	Re-check the readings (RI usually has three decimals, SG two) and the optic character. The stone may also be outside the system's scope; seek a laboratory test.
Changes to the program files do not appear	In the Prolog terminal type <code>make.</code> to reload changed files, then refresh the browser (Ctrl+F5). Or stop and restart the system.
Windows Firewall message on first start	Choose Allow for private networks.
The page fonts look plain	You are offline; the page falls back to standard fonts. Everything still works.

13. Console mode (without a browser)

If the web interface cannot be used, the same functions are available in the terminal.

1. Start SWI-Prolog with `swipl main.pl`.
 2. Type `start.` A menu appears:
3. What would you like to do?
 4. 1. Identify a stone - "I don't know what it is" (forward chaining)
 5. 2. Check a claim - "The seller says it's a ruby" (backward chaining)
 6. 3. Step by step - asks only what it needs; type WHY at any question
 7. 4. Ask in English - e.g. is it a ruby if the RI is 1.765 and it is red

8. 5. Run the demo - natural blue sapphire example
9. 6. Quit

Choose 1-6:

10. Type a number and press Enter (no full stop is needed inside the menu). Answer each question by typing a number or a value; press Enter on its own if you do not know.

11. In step-by-step mode, type `why` at any question to see why it is being asked:

```
12. What is the optic character?
13.    0. Don't know
14.    1. Isotropic
15.    2. Uniaxial
16.    ...
17. > why
18. WHY I AM ASKING:
19. - I am trying to find out: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND
    RI in
20. Corundum range AND SG in Corundum range THEN Species: Corundum.
```

- To test rule r1 I need to know: the optic character.

21. After each result you can type `y` to see the technical trace.

22. Choose **6** to leave the menu, then type `halt`.

The command `demo` runs the worked blue sapphire example directly.

14. Running the automated tests

1. Start SWI-Prolog with `swipl main.pl`.
2. Type `run_tests`.
3. The last line should read:

```
% All 41 tests passed
```

The rule base can also be exported to a spreadsheet file with `export_rules_csv('rules_table.csv')`.

15. Quick reference

Table 5. Commands

Command (type at the <code>?-</code> prompt)	Effect
<code>swipl main.pl</code>	(in the terminal) Load the system.
<code>server(8080).</code>	Start the web interface at <code>http://localhost:8080</code> .
<code>stop_server(8080).</code>	Stop the web interface.

start.	Start the console interface.
demo.	Run the worked blue sapphire example.
run_tests.	Run the 41 automated tests.
make.	Reload program files after editing them.
halt.	Leave SWI-Prolog.

Table 6. Web pages

Address	Page
http://localhost:8080/	Home page, form, examples, glossary
http://localhost:8080/consult	Step-by-step consultation
http://localhost:8080/rules	Knowledge base: gem facts and all rules